CHAPTER 10

Serial and I2C

In the last chapter, we looked at how packages of commonly used code called libraries can be useful for working with a range of different kinds of hardware. Usually, these libraries are written for very specific devices or hardware that uses particular components. In this chapter, we will look at a few more libraries for working with more standardized communication protocols that can be used to communicate between many different types of devices that speak the same language. In Chapter 6 we got a taste for using hardware serial communications for monitoring an analog input, although this was a fairly basic one-way communication.

We will revisit hardware serial in this chapter, looking at different functions that can be used to not only send but receive information as well, using Project 9 Serial to Servo, for our discussion. We will then move on to serial communications using a software library instead of the Arduino's built-in hardware to expand the number of things that we can talk to with serial, including the tag reader used in Project 10 RFID Reader. After that, we should have a look at another useful serial communication in the form of the Inter-Integrated Circuit (I2C) data bus, useful for talking with other hardware devices like sensors, displays, and the real-time clock used in Project 11 Serial Time Clock.

What's needed for this chapter:

- Arduino Uno
- Hobby servo (Hitec HS-322 or similar)
- Innovations ID-12 or ID-20 RFID reader and breakout board
- Assorted 125kHz RFID tags
- Piezoelectric speaker or LED and appropriate 220-ohm resistor
- DS1307 real time clock breakout board (SparkFun or Adafruit both make these)
- Hookup wires
- Solderless breadboard

Using Hardware Serial

Every Arduino has at least one hardware serial port. We have already used this port to send our compiled sketches to the Arduino board each time we upload our code. The single hardware serial port on the Arduino Uno is connected to both the USB to Serial convertor chip on the interface board, as well as